



MOHAMMAD REZA NASIRZADEH

PERSONAL DATA

NAME: Mohammad Reza Nasirzadeh
PLACE AND DATE OF BIRTH: Yazd, Iran | 11 January 1995
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WORK EXPERIENCE

06/2020 - 04/2023 | Research assistant at Institute For Research In Fundamental Sciences (IPM), Prof. F. S. Tabatabaei

SCIENTIFIC EDUCATION

06/2019-03/2022 | M.SC. IN GRAVITY & COSMOLOGY
University of Kharazmi

09/2014-06/2019 | B.SC. IN MOLECULAR PHYSICS
University of Yazd

RESEARCH INTERESTS

- Multi-wavelength Astronomy (Radio/Infrared/Optical)
- Interstellar Medium Physics and Galactic Structure
- Cosmic Ray Propagation in Galactic Environments
- Star Formation in Galactic and Extra-galactic Environments
- Thermal/Non-Thermal Processes in Stellar and Galactic Systems
- Molecular Gas Dynamics and Dust Properties in Stellar Formation
- Magnetic Fields in Star Formation Regions and Stellar Evolution
- Faraday Tomography and Polarization Studies
- Stellar Populations and Galactic Evolution
- Large Survey Data Analysis and Statistical Methods

RESEARCH EXPERIENCES

Current Project:

- Exploring Magnetic Fields in Molecular Clouds through Faraday Rotation and Faraday Tomography
- Under guidance of: Dr. Mehrnoosh Tahani, Assistant Professor, University of South Carolina.

Related Collaboration: **GMIMS (Global Magneto-Ionic Medium Survey)** international collaboration.

M.Sc.:

- Research Assistant, Studying Radio-IR Correlation in IC 342 based on LoTSS Data
 - School of Astronomy, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran ([Home page](#))
 - Supervisor: Dr. F. S. Tabatabaei, School of Astronomy, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran
- Related Collaboration: **LOFAR Key Science Projects (Magnetic Key Science Project), NEARBY GALAXIES group.**

B.Sc.:

- Analysis and Reduction of Data Taken from the Giant Meteor Wave Radio Telescope (GMRT)
- Research on Phenomenon related to the Very Low Frequency (VLF) Band of Radio Electromagnetism Spectrum, including:
 - * Solar Flares and Sudden Ionospheric Disturbances (SIDs)
 - * Meteor
 - * Meteorological Phenomenon
 - * Ionospheric Anomalies
 - * Earthquake Prediction
- Designing and Constructing Small Radio Detectors to Study the Phenomenon in VLF Band
- Supervisor: Dr. Zahra. Rezaei, Yazd University, Yazd, Iran

PUBLICATION

- **Nasirzadeh, M.R.**, Tabatabaei, F.S., Beck, R., Heesen, V., Howaida, P., Reina-Campos, M., Paladino, R., Dettmar, R.-J., Chyzy, K.T. Astronomy & Astrophysics Journal, [Radio-FIR correlation: A probe into cosmic ray propagation in the nearby galaxy IC 342.](#)

OBSERVATIONAL DATA EXPERIENCE

- Radio instruments and surveys: LOFAR, GMIMS, CHIME, and VLA.
- Planck's dust polarization data analysis.
- UV data processing with GALEX.
- IR/FIR analysis using Spitzer, Herschel, and WISE.
- 3D interstellar extinction Profile analysis.
- HI spectral line data analysis.
- + Other dataset (H-alpha/Gamma/Extinction/CO)

TALK AND PRESENTATIONS

- **Nasirzadeh, M.R.**, Tabatabaei, F.S., Heesen, V., Beck, R. (2024). Radio-FIR correlation: A probe into cosmic ray propagation in IC 342 (Poster). Cosmic Magnetism with Radio Astronomy 2024
- **Nasirzadeh, M.R.**, Tabatabaei, F.S., Heesen, V., Beck, R. (2023). Radio-IR Correlation in IC342. 16th National Conference on Astronomy and Astrophysics of Iran.
- Howaida, P., Tabatabaei, F.S., **Nasirzadeh, M.R.**, Heesen, V., Shimwell, T.W. (2023). LOFAR LoTSS Resolved Radio-IR Correlation in Nearby Galaxies (Poster). SPARCS XI (The SKA Pathfinder Radio Continuum Survey) Meeting.
- **Nasirzadeh, M.R.**, Tabatabaei, F.S., Heesen, V., Beck, R. (2021). A detailed study of the Radio-IR correlation in IC342 (Wavelet and thermal-non thermal separation). Annual Meeting of the LOFAR Magnetism Key Science Project (MKSP).
- **Nasirzadeh, M.R.**, Tabatabaei, F.S. (2021). Magnetic Fields in IC 342. OAC Talk at Institute for Research in Fundamental Sciences (IPM).
- **Nasirzadeh, M.R.**, Tabatabaei, F.S., Heesen, V., Beck, R. (2020). Study of the Radio-IR correlation in IC342. Annual Meeting of the LOFAR Magnetism Key Science Project (MKSP).
- **Nasirzadeh, M.R.**, Tabatabaei, F.S., Heesen, V., Beck, R. (2020). A Detailed Study of Radio-IR Correlation in IC 342 based on LoTSS Data. Research Seminar School of Astronomy (IPM).

SELECTED COMPUTER SKILLS

Operating Systems	Windows, Linux
Analysis & Programming	AIPS, DS9, Python (NumPy, AstroPy and...)

WORKSHOP AND SEMINAR ATTENDED

- 2025, Tehran, Iran | A Computational Look at Astrophysical Flows
- 2024, Pisa, Italy | Cosmic Magnetism with Radio Astronomy (COMRAD2024)
- 2022, Tehran, Iran, IPM | Transient events and Multi-Messenger Astrophysics
- 2022, Tehran, Iran, IPM | Workshop on Formation, Evolution and Structure of Galaxies
- 2021, Byurakan, Armenia | Second Regional Astronomical Summer School (2RASS)
- 2021, Tehran, Iran, IPM | Radio Astronomy Workshop (RAW) “Observation and Data Reduction”
2021 | Annual meeting of the LOFAR Magnetism Key Science Project (MKSP) - 2021
- 2020 (Virtual) | Annual Meeting of the Astronomische Gesellschaft 2020 Virtual meeting
- 2020 (Virtual) | Annual meeting of the LOFAR Magnetism Key Science Project (MKSP) - 2020
- 2020, Zanjan, Iran, IASBS | 23rd Winter School on Physics
- 2019, Tehran, Iran, IPM | Baryons in Galaxies and Beyond.
- 2018, Tehran, Iran, SBU | Workshop on Machine Learning in Physics,
- 2017, Tehran, Iran, IPM | IPM Radio Astronomy School
- 2016, Yazd, Iran, Yazd University | French- Iranian Workshop on the Mercury Transit of 2016
- 2015, Yazd, Iran, Yazd University | 1st Workshop on Radio Astronomy- Interferometry and Data Analysis in Iran

SCIENCE OUTREACH

- Voluntary Astronomy, Geography and History teacher for refugee children (2019).
- Voluntary Literature teacher for refugee children (2018-2019).
- Voluntary Scientific editor for different independent Astronomy and Science groups (2016-2022)
- Head of Student Astronomical Society of Yazd University (2017-2018).

SCHOLARSHIPS, HONOURS AND AWARDS

- IPM annual grant in Astronomy and Astrophysics for research project “Detailed Study of Radio-IR correlation in IC 342” (2019-2022).
- Top 10% in MSc University exam entrance (2019).
- Top 5% among Gravity And Cosmology MSc students at Kharazmi University (2019-2022).

REFERENCE

- [Prof. F. S .Tabatabaei](#): Professor, School of Astronomy, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran, +982123105069
Email: ftaba@ipm.ir
- [Dr. Mehrnoosh Tahani](#): Assistant Professor, University of South Carolina
Email: mtahani@mailbox.sc.edu